Composite Rectilinear Figures: Area and Perimeter Answer Key

Grade 3 — Measurement and Data

Quick Answers

1. 36	4. 30	7. 75	10. 8
2. 34	5. 54	8. 126	11. 96
3. 47	6. 7	9. 56	12. 72

Every answer below is written the way a third grader should say it out loud: *what the shape was cut into*, then *the numbers*. Accept any correct decomposition — a shape can be cut into rectangles in more than one way, and the total must come out the same.

1. Sticker, area.

Solution: Cut across at the height of 3: a bottom rectangle 8 by 3 and a top rectangle 4 by 3.

$$\begin{aligned} \text{bottom} &= 8 \times 3 = 24 \\ \text{top} &= 4 \times 3 = 12 \\ \text{total} &= 24 + 12 = 36 \end{aligned}$$

$\boxed{36 \text{ cm}^2}$

Check the other way: the whole 8×6 rectangle is 48, minus the missing $4 \times 3 = 12$ corner, which is also 36.

2. Ribbon, perimeter.

Solution: Walk all the way around and add every side, without skipping the two short sides inside the notch:

$$10 + 4 + 5 + 3 + 5 + 7 = 34$$

$\boxed{34 \text{ cm}}$

Watch for: students who add only the four "outside" sides $(10 + 4 + 5 + 7)$ get 26 and have missed two sides.

3. Two rectangles, area.

Solution: Cut across at the height of 3: a bottom rectangle 9 by 3, and an upper right rectangle 5 wide by 4 tall.

$$\begin{aligned} \text{bottom} &= 9 \times 3 = 27 \\ \text{upper right} &= 5 \times 4 = 20 \\ \text{total} &= 27 + 20 = 47 \end{aligned}$$

$\boxed{47 \text{ cm}^2}$

4. Name tag, perimeter.

Solution: There are eight sides. Add them in order around the shape:

$$4 + 4 + 2 + 3 + 8 + 3 + 2 + 4 = 30$$

$\boxed{30 \text{ cm}}$

Note: area is not asked for here. A student who multiplies has answered a different question.

5. Maya's way (subtracting), area.

Solution: The whole rectangle would be 10 across and 6 tall. The bite taken out of the corner is 3 across and 2 tall.

$$\begin{aligned} \text{whole rectangle} &= 10 \times 6 = 60 \\ \text{missing corner} &= 3 \times 2 = 6 \\ \text{shape} &= 60 - 6 = 54 \end{aligned}$$

$\boxed{54 \text{ cm}^2}$

Check by adding instead: $10 \times 4 = 40$ and $7 \times 2 = 14$, and $40 + 14$ is the same total.

6. Missing side.

Solution: The two top pieces lie along the same straight line as the bottom, so together they must be as long as the bottom.

$$5 + ? = 12 \qquad \Rightarrow \qquad ? = 12 - 5 = 7$$

$\boxed{7 \text{ cm}}$

7. Rug, area.

Solution: Cut across at the height of 5: a bottom rectangle 12 by 5 and a top rectangle 5 by 3.

$$\begin{aligned} \text{bottom} &= 12 \times 5 = 60 \\ \text{top} &= 5 \times 3 = 15 \\ \text{total} &= 60 + 15 = 75 \end{aligned}$$

$\boxed{75 \text{ ft}^2}$

8. Patio with a flower bed, area.

Solution: Start with the whole rectangle and take away the flower bed.

$$\begin{aligned} \text{whole patio} &= 15 \times 10 = 150 \\ \text{flower bed} &= 4 \times 6 = 24 \\ \text{paved part} &= 150 - 24 = 126 \end{aligned}$$

$\boxed{126 \text{ ft}^2}$

9. Garden fence, perimeter.

Solution: Add the six sides all the way around:

$$16 + 5 + 9 + 7 + 7 + 12 = 56$$

$\boxed{56 \text{ ft}}$

Nice pattern to point out: the two horizontal pieces (9 and 7) add to the bottom (16), and the two vertical pieces (5 and 7) add to the left side (12) — so this perimeter is the same as a 16 by 12 rectangle's.

10. Missing side on the deck.

Solution: The right-hand side and the short piece above it stack up to make the whole left side, because both go from the bottom edge to the top edge.

$$? + 5 = 13 \qquad \Rightarrow \qquad ? = 13 - 5 = 8$$

$\boxed{8 \text{ ft}}$

11. Challenge — porch floor, area.

Solution: First the unlabelled side. The left side is 9 and the right side is 6, so the short vertical piece is $9 - 6 = 3$. Now take the whole 12 by 9 rectangle and subtract the missing corner, which is 4 across by 3 tall.

$$\begin{aligned} \text{whole rectangle} &= 12 \times 9 = 108 \\ \text{missing corner} &= 4 \times 3 = 12 \\ \text{porch} &= 108 - 12 = 96 \end{aligned}$$

$\boxed{96 \text{ ft}^2}$

Adding instead: $12 \times 6 = 72$ plus $8 \times 3 = 24$ gives the same total.

12. Challenge — three steps, area.

Solution: Cut the shape into three flat steps, each 3 tall:

$$\begin{aligned} \text{bottom step} &= 12 \times 3 = 36 \\ \text{middle step} &= 8 \times 3 = 24 \\ \text{top step} &= 4 \times 3 = 12 \\ \text{total} &= 36 + 24 + 12 = 72 \end{aligned}$$

$\boxed{72 \text{ ft}^2}$

Watch for: a student who uses 12, 8 and 4 as the *heights* as well as the widths has cut the shape into overlapping pieces — have them shade each step before multiplying.